IEEE 4th International Conference on Universal Village · UV2018 · Session 9b-6

Video-based Fall Detection for Seniors with Human Pose Estimation

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***Abstract*—In recent years, aging of population and empty nest problem are becoming more and more severe. In addition, fall is the leading cause of death for seniors both in China and the U.S. Therefore, automatic fall detection for seniors is required in smart home and smart healthcare system. Currently, for its convenience and low cost, video-based method is the optimal method compared with other methods such as wearable sensor and ambient sensor in the field of indoor fall detection. In this paper, we propose a novel 2D video-based fall detection pipeline with human pose estimation. Firstly, we used OpenPose to extract the positions of human joints in raw data. Secondly, these data with augmented features became the input of a convolution neural network so that we can extract multi-layered features. Thirdly, a binary classification was conducted through neural network. For comparison, we also used SVM as the classifier. At last, we achieved relatively high sensitivity and specificity when compared our results with other state-of-the-art approaches on three public fall datasets.**

Keywords—Fall Detection, Human Pose Estimation, Smart Healthcare

I. INTRODUCTION

Fall detection is an active research area under the field of smart home and smart healthcare system. According to National Bureau of Statistics of China [1], there are more than 158 million seniors in China, and this number will be larger in the next few years. In addition, there are more than half of them are living alone, whom are called "empty nest seniors". Worse still, fall is a frequently happened event. Concretely, one-third of seniors fall down each year [2]. Fall is not a big deal for teenagers, but is very dangerous for seniors because it may cause some severe disease such as hip fractures, head injury and heart attack. Thus and unsurprisingly, fall is the leading cause of death in both China and the United States [3]. Fortunately, there is an obvious positive correlation between the arrival time of medical treatment and death rate. In other words, if medical treatment can arrive in a very short time, those who fall down will probably have no serious consequences. In reality, many seniors live alone so if they fall down, their families cannot find out. However, with the alarm system which can automatically send messages when falls are detected, their families or hospitals can react quickly to save lives or reduce the severity of injuries.

Currently, there are three methods for fall detection, including wearable sensor, ambient sensor and video-based detection.

Wearable sensor: Chen *et al.* [4] and Bourke *et al.* [5] proposed a fall detection method using tri-axial acceleration sensor which can be worn by seniors. Wearable sensor can provide high sensitivity and specificity while it also has some drawbacks. Seniors may forget to wear their sensors or forget to recharge. Take Apple Watch Series 4 as an example, it can do fall detection task. However, it should be recharged every 18 hours according to its official website.

Ambient sensor: M. Alwan *et al.* [6] introduced a floor vibration-based fall detector which is completely passive and unobtrusive to seniors. Although the accuracy is high, the price of those detectors are unaffordable for normal families. Therefore, we need other methods for fall detection that are cost-effective.

Video-based detection: Video-based method is a reliable detection method with low cost, and seniors do not need to wear or recharge the cameras. But it also has some disadvantages. For example, some seniors are uncomfortable of the feeling of being watched. In general, video-based method is the optimal for its high accuracy and low cost.

II. RELATED WORKS

After probing into video-based fall detection method, we found that there are two prevalent cameras are used for video capturing. One is RGB camera, the other is depth camera,

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which can generate 3D images. Since 3D images contain depth features, many researchers prefer using depth camera for fall detection. Zambanini *et al.* [7] and Rougier *et al.* [8] introduce fall detection algorithms based on 3D cameras. More specifically, Stone *et al.* [9] achieved a high recall rate using a Microsoft Kinect. However, 2D RGB camera may be better than 3D depth camera for several reasons. First, the visual range of depth camera is shorter than 2D camera. For example, Kinect can only detect human motion from 0.4 meters to 3 meters while most of the 2D camera are able to detect more than 3 meters. Second, the visual field of depth camera is about 50 degrees in horizontal and vertical directions, which is far from enough in actual fall detection. In contrast, most 2D cameras have visual field more than 90 degrees. Third and most importantly, the cost of RGB camera is more reasonable, and most of the indoor surveillance system are composed by 2D cameras nowadays. Therefore, fall detection based on 2D camera can be applied on many environments with lower cost. In summary, 2D RGB camera performs better, and most of the latest researches are based on 2D camera.

Chen *et al.* [10] applied foreground extraction from 2D images, and used human shape as a main feature to distinguish fall-down incident. With too few fall-down examples, we can not comprehensively evaluate their algorithms. In 2016, Wang and al. [11] introduced a novel pipeline, and divided fall detection into two steps. First, they classified each frame by SVM, and labeled them. Second, they used label sequence as the input of SVM, and classified the whole video. PCANet plays a role as a feature extractor. This method was tested on a public dataset. They achieved 89.2% Sensitivity and 90.3% Specificity. After a year, Wang and Kun *et al.* [12] combined Histograms of Oriented(HOG), Local Binary Pattern(LBP) and feature extracted by the Deep Learning Framework Caffe to form a new augmented feature. Then, they used this combined feature for fall detection in their formal pipeline on their previews paper [11]. This augmented feature actually improves their results, and reached 93.7% Sensitivity and 92.0% Specificity. In 2018, Zerrouki *et al.* [13] introduced a SVM-Hidden Markov Models as a combined classifier, since they think fall-down event is related to time series. They only got a overall accuracy on a public dataset. However, in this case, we highly concern about the Sensitivity, which is the recall rate of positive samples, instead of Specificity or Average Accuracy. This is because that in fall-down event, even if our algorithm successfully finds all the negative (non-fall) samples, and achieve 100% Specificity, if we can not prove its ability on finding positive samples, it is still a useless algorithm.

Consequently, there is huge space for improvement in fall detection, and our approach can improve the performance of video-based fall detection for seniors with the novel augmented feature that is human post data.

III. Our Approach

In our approach of fall detection, we build a novel pipeline that is showed by figure1.

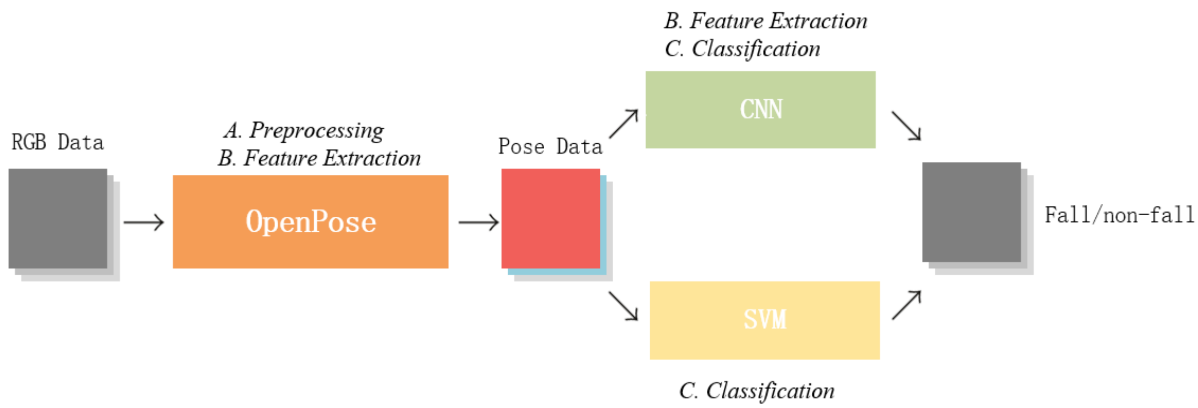

Fig. 1. The main pipeline of fall detection with human pose estimation

A. Preprocessing

In many machine learning tasks, preprocessing is one of the most important steps, and so is fall detection. We may find many researchers [11] [12] improved a little bit in this part. In our approach, OpenPose [14] plays an important role not only in preprocessing steps, but also in feature extraction steps. OpenPose was built by Perceptual Computing Lab at CMU, and it can detect human key points in images (2D and 3D) for multi-person. In 2D images detection, its core algorithm is from Cao et al. [15]. Their method, which is called greedy bottom-up parsing, can efficiently find joints and facial key points of multi-person in images (FPS performance is related to the platform). Then, Part Affinity Fields(PAFs) helps to connect body parts with individuals.

In our method, after preprocessing, pose data can be generated. Concretely, pose data contains two parts. One is images with pose keypoints, the other is pure keypoints' coordinates as array class.

B. Feature Extraction

Feature extraction is highly related to the whole performance of the fall detection system. Previously, array classes with keypoints coordinates are extracted by OpenPose, which is a feature extraction step. However, the images with pose keypoints are just data with augmented features. Therefore, these images should be further processed.

In our approach, a convolutional neural network is used as a feature extractor. Concretely, we applied VGG-16 [16] on these images sequences with pose data because its performance is better. More specific, each sequence contains 10-20 frames according to previous researches [10]. Then, multi-level features can be extracted for following classification.

C. Binary Classification

After preprocessing and feature extraction, the pipeline comes to the last step, binary classification. Since we have pure keypoints' coordinates, a SVM with Gaussian Kernel can utilize them as input features. Concretely, 15 body features are used in each image. Although OpenPose can general 18 or 25 keypoints for 2D images, those redundant keypoints are mostly facial or finger keypoints, which are not important in fall detection, and will slow down the processing steps.

Apart from support vector machine, CNN can also do classification tasks. There are 13 convolutional layers and 3 fully connected layers in VGG-16 architecture. We fixed other layers of trained neural networks, then used fall dataset to train the last fully connected layer, which is called transfer learning [17]. The soft-max function performs well in this binary classification task.

After all these works, there are two results of each sequence of images, positive or negative. For improving sensitivity, we decided to take union of positive samples. In other words, each sample in test set will be labeled as positive if any one of two classification models thinks it is a positive sample. Then other samples will be labeled as negative samples.

IV. Experiment

A. Platform

We used normal personal computers with Windows 10 operating system to conduct this experiment. Specifically, the laptop contains an Intel i7 CPU, a Nvidia GTX GPU and an 8G RAM.

B. Datasets

In order to test the generalization performance of our approach, we found three public datasets which has been widely used in fall detection field.

Multiple Cameras Fall Dataset [18] contains 24 scenarios recorded with 8 video cameras from different perspective in a room. More specific, the first 22 scenarios contain a fall and confounding events, the last 2 ones contain only confounding events.

UR Fall Detection Dataset [19] contains 30 falls and 40 confounding activities of daily life sequences. Fall events are recorded by 2 Microsoft Kinect cameras while non-fall events are recorded by only one device and an accelerometer.

The Fall Detection Dataset [20] is a dataset in realistic video surveillance data recorded by a single camera. With 25 FPS and 320x240 resolution, this dataset contains 191 videos and labels representing the fall-down position in the video sequence. This dataset is characterized by its multiple scenarios, including home, coffee room, office and lecture room, which can better test the robustness of fall detection algorithms

C. Results

We then applied those three steps on these three fall datasets. Figure 2,3 and 4 are the sample results of Multiple Cameras Fall Dataset, UR Fall Detection Dataset and The Fall Detection Dataset.

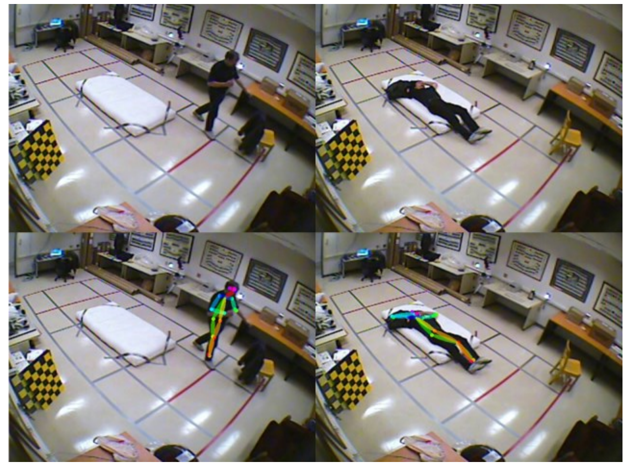

Fig. 2. Sample results of Multiple Cameras Fall Dataset

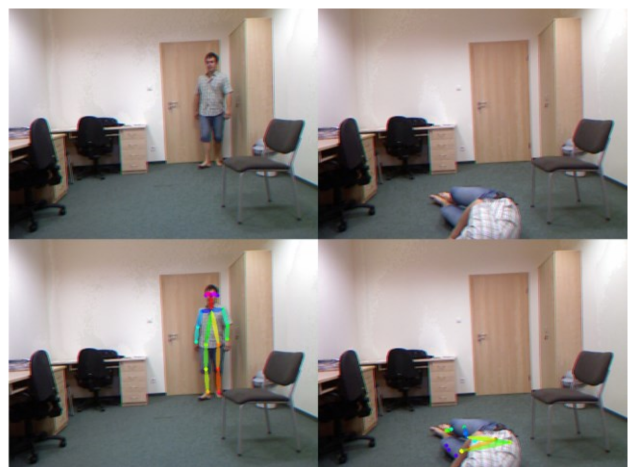

Fig. 3. Sample results of UR Fall Detection Dataset

Fig. 4. Sample results of UR Fall Detection Dataset

Due to time and place limitation, we did not put the classification experiment on this paper. To whom it may interested, this part of experiment can be conducted according to our pipeline.

As we can see, the 15 joint keypoints of each person in three datasets can be correctly represented. Although the images sequence of these datasets only contain one person, we can detection more people in a room. When people fall down, their body keypoints begin to distort dramatically, and their poses are very unnatural. Actually, these features are very clear in each video sequence (10-20 images). Moreover, static things such as furniture can be blocked. Therefore, it can greatly improve the performance of feature extraction and binary classification.

V. Conclusions

In conclusion, this paper proposes a novel 2D video-based fall detection pipeline with human pose estimation as the method of feature augmentation. Concretely, we first used OpenPose to extract the coordination of human body keypoints in raw RGB data. Secondly, these data with augmented features became the input of a convolution neural network so that we can extract multi-layered features. Thirdly, a binary classification was conducted through neural network. For comparison, we also used SVM as the classifier. At last, we achieved relatively high sensitivity and specificity when compared our results with other state-of-the-art approaches on three public fall datasets.

In the future, researchers may conduct more experiment of fall detection in dark environment at night and outdoor environment. Since the difficulty of fall detection in those environment, currently no many papers discussed those situations.

Acknowledgment

We greatly acknowledge the help from all members of the Universal Village Group at MIT. With their innovative thinking and great suggestions, our research could conduct more efficiently.

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